Project Description

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House Price Predictions via the Machine Learning Models of Linear Regression and KNN

In this project, I use two machine learning models and three pathways (Python, RapidMiner, and MongoDB) to study a housing dataset. I employ the dataset and the machine learning techniques of linear regression and k-Nearest-Neighbor to generate two separate models that predict house price using other features in the dataset. House prices are determined by many factors, but I will assess whether a few of the available variables are useful as a rough prediction guide.

Python

In Python, I imported the necessary libraries and the dataset, conducted an exploratory data analysis, and showed several data visualizations. The first machine learning model I constructed used linear regression and tested how well the Number of Bedrooms, the Number of Bathrooms, and the Square Feet of Living Space predicted house Price. I obtained the Mean Squared Error and the R squared score, and plotted a visualization of the results versus the predictions. I next constructed a KNN model testing the same three variables on the Price target variable. I determined this model's Mean Squared Error and R squared score, and plotted a visualization of the results, plus a residual plot.

RapidMiner

I used the RapidMiner data platform to construct the same Linear Regression model that I had in Python above. Next, I generated a correlation matrix of the selected variables.

MongoDB

I loaded the dataset into the MongoDB data platform and conducted basic queries and a Pandas dataframe display.

Conclusions

Both the Linear Regression model and the KNN model had utility in predicting house price given the number of bedrooms, number of bathrooms, and the square feet of living area. Note that I did not choose all of the variables given in the dataset. Instead, I constructed an individual consumer preference-driven model. If a prospective buyer has heightened interest in the number of bed/bathrooms and living space, this model gives an indicator of how much these preferences will predict price, all other features notwithstanding. One ease-of-use feature of this model is that the three attributes listed are easy to obtain in real estate listings. Numerical statistic results are discussed in each project file.